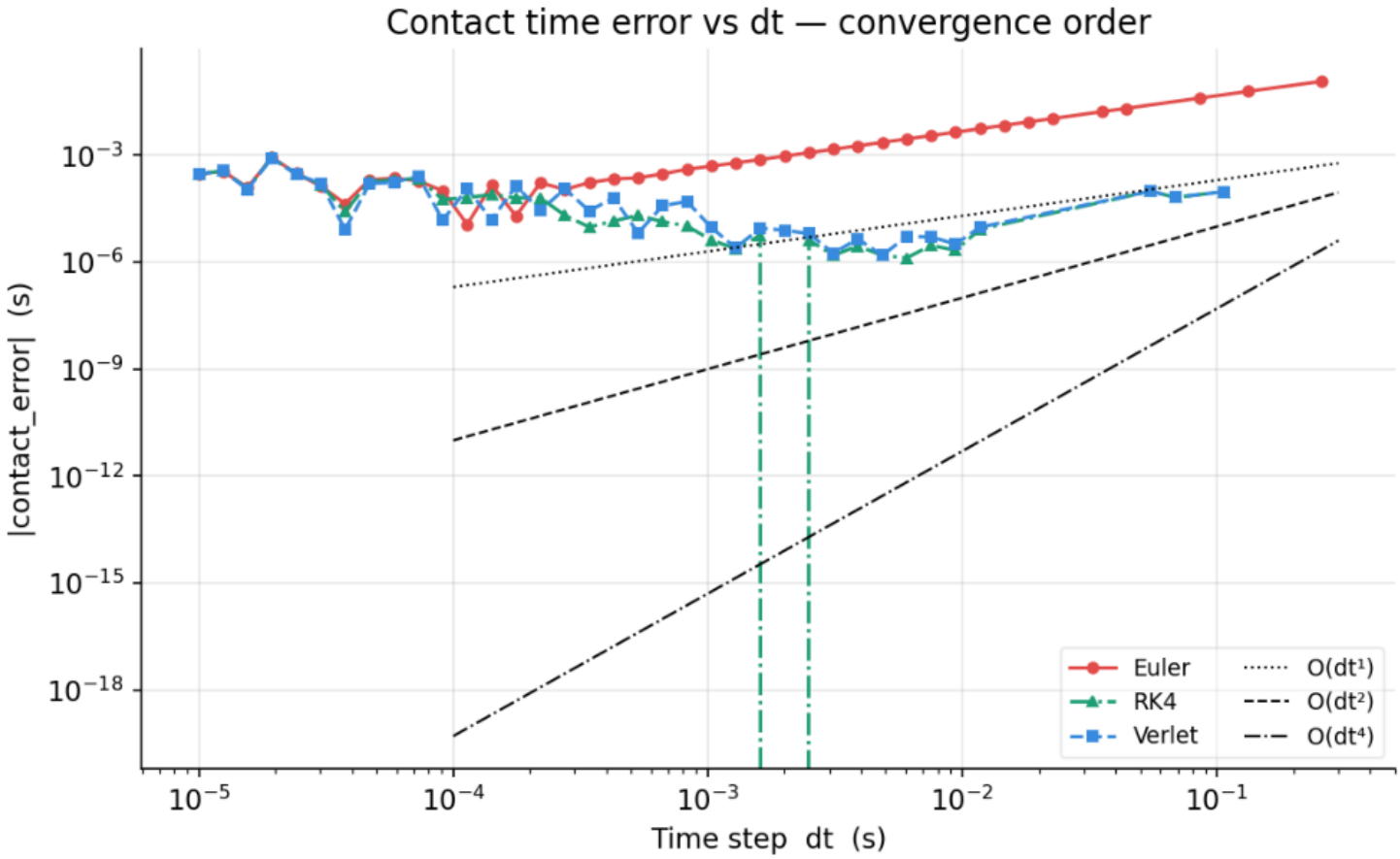
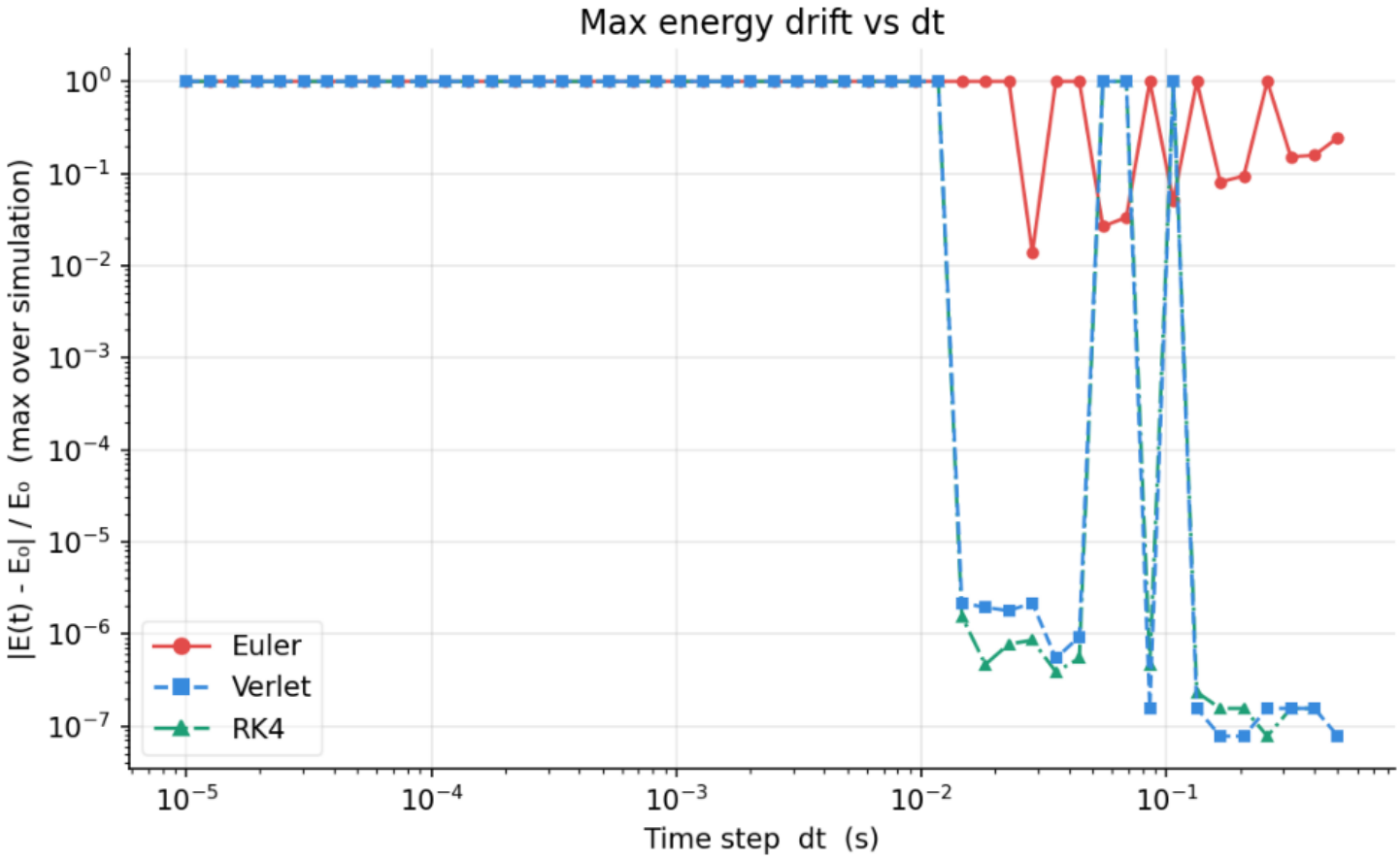
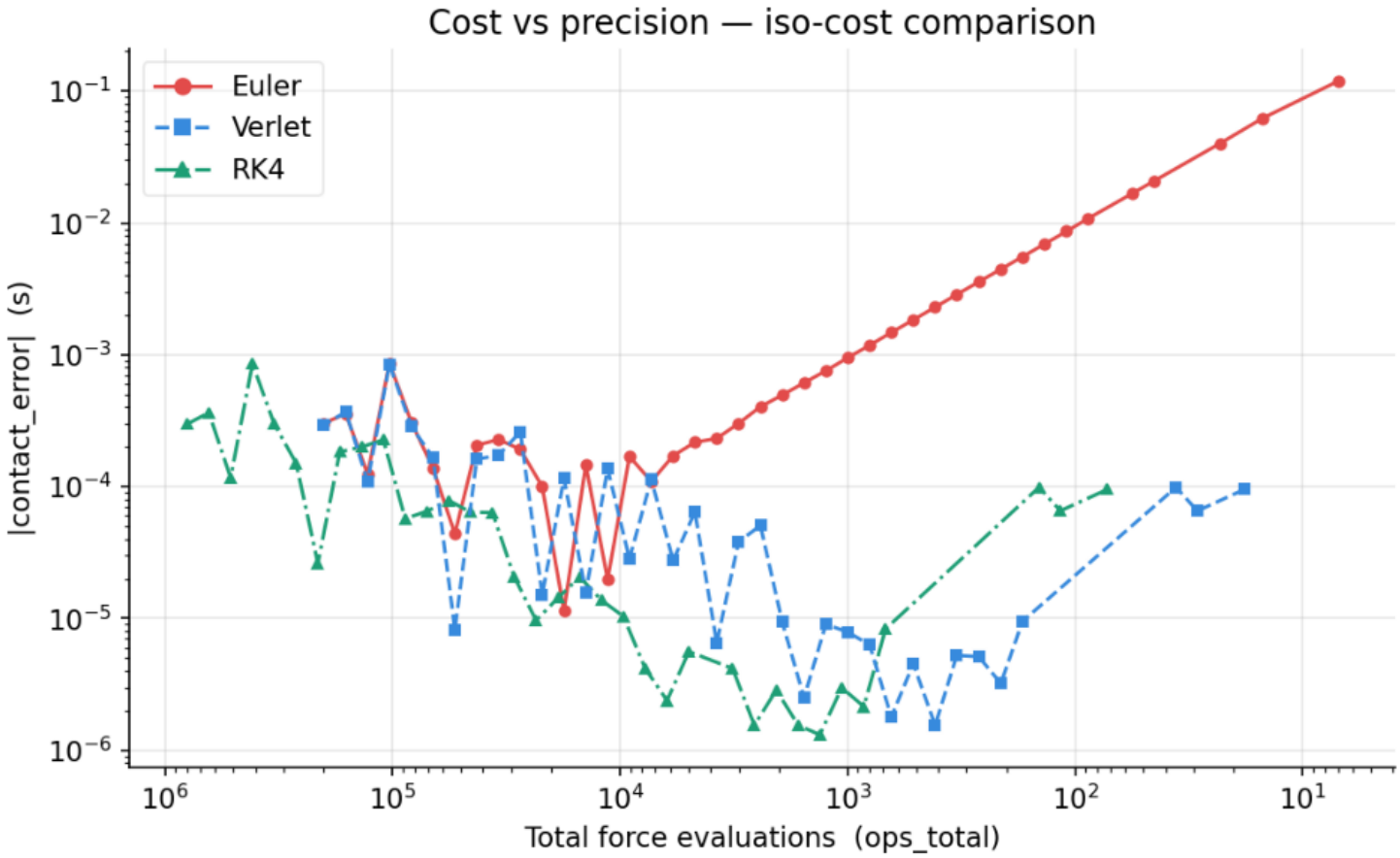
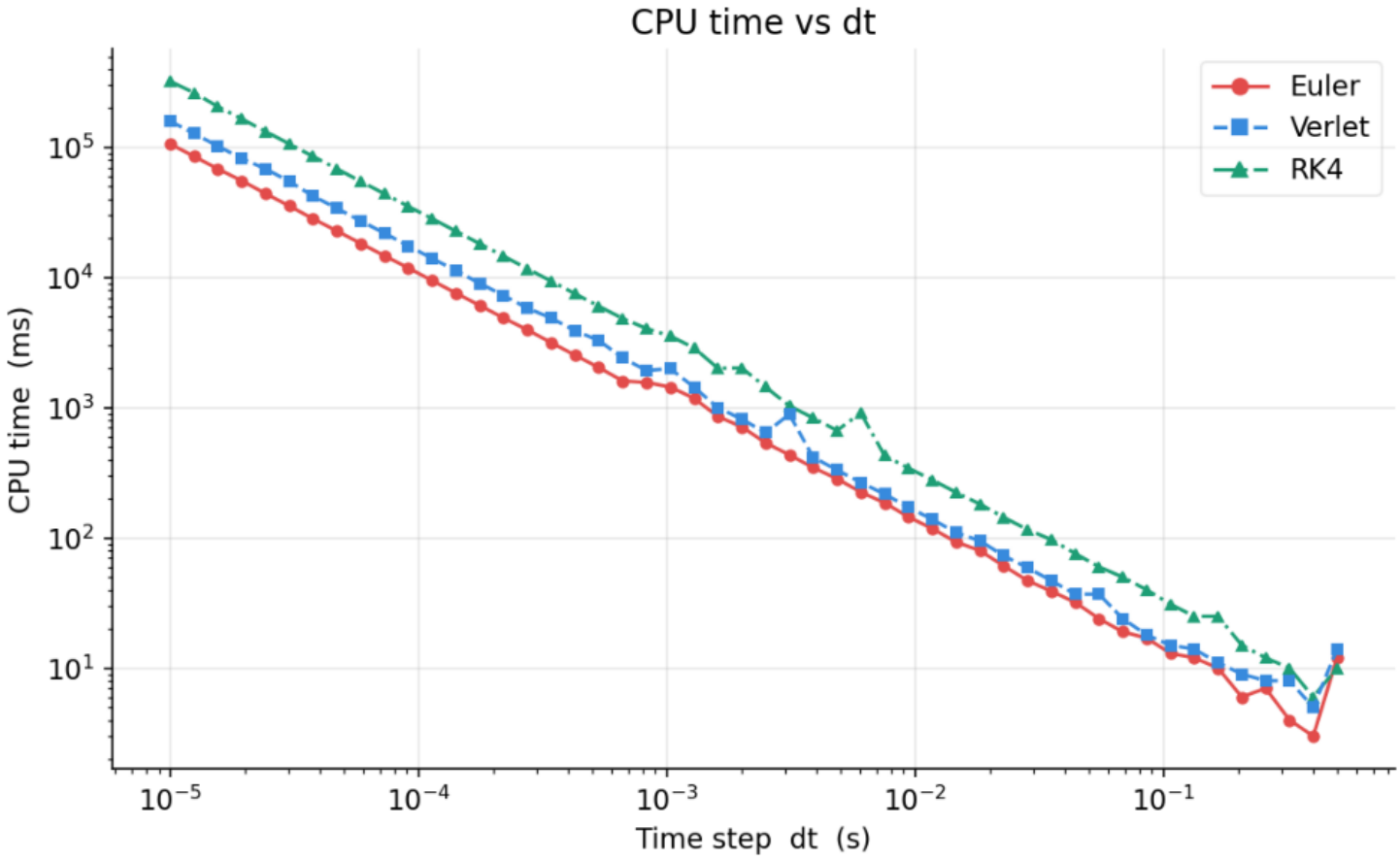
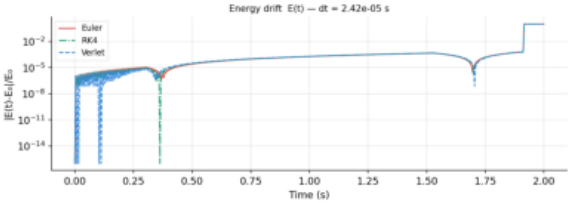
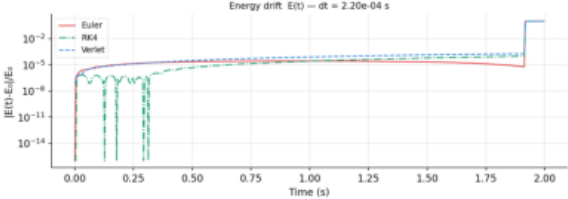

Contact time error vs dt — convergence order
|contact_error| (s)
Time step dt (s)
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
Euler
RK4
Verlet
O(dt¹)
O(dt²)
O(dt⁴)

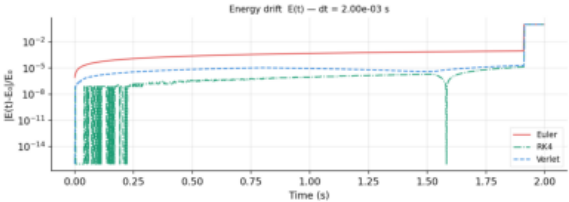

Max energy drift vs dt
|E(t) - E₀| / E₀ (max over simulation)
Time step dt (s)
Euler
Verlet
RK4
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}

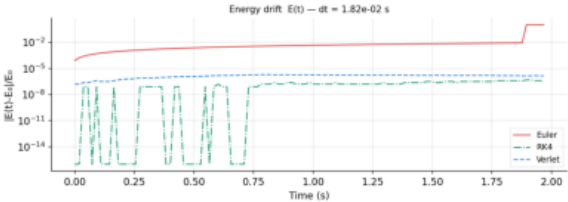

Cost vs precision — iso-cost comparison
Euler
Verlet
RK4
|contact_error| (s)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
Total force evaluations (ops_total)

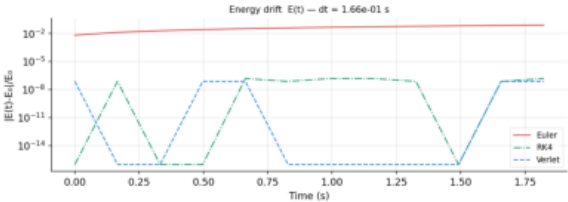

CPU time vs dt
Euler
Verlet
RK4
CPU time (ms)
10^5
10^4
10^3
10^2
10^1
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
Time step dt (s)


Energy drift E(t) — dt = 2.42e-05 s
Euler
RK4
Verlet
|E(t)-E₀|/E₀
Time (s)


Energy drift E(t) — dt = 2.20e-04 s
Euler
RK4
Verlet
|E(t)-E₀|/E₀
Time (s)


Energy drift E(t) — dt = 2.00e-03 s
Euler
RK4
Verlet
|E(t)-E₀|/E₀
Time (s)


Energy drift E(t) — dt = 1.82e-02 s
Euler
RK4
Verlet
|E(t)-E₀|/E₀
Time (s)


Energy drift E(t) — dt = 1.66e-01 s
Euler
RK4
Verlet
|E(t)-E₀|/E₀
Time (s)